

Project Name: TaskFlow, SaaS Project Management Tool

Problem

Small teams often struggle to manage projects using scattered tools like chat apps, spreadsheets, and email. Tasks get lost, deadlines are missed, and there is no clear visibility of who is doing what.

Most powerful project management tools are either too complex or too expensive for small teams and startups.

Solution

TaskFlow is a web based SaaS project management platform built with the MERN stack. It helps teams organize work using boards, lists, and tasks in a simple visual interface.

It supports collaboration in realtime, team roles, file sharing, and progress tracking, all inside one workspace.

The goal is to provide a lightweight but powerful system that improves productivity and team communication.

Target Users

- Startup teams
- Student project groups
- Remote teams
- Freelancers working with clients

Key Value

- Clear task ownership
- Visual project tracking
- Realtime collaboration
- Central place for files, comments, and deadlines

Tech Stack

Frontend: React, Context API or Redux, Tailwind CSS

Backend: Node.js, Express

Database: MongoDB

Auth: JWT

Realtime: Socket.io

File Storage: Cloudinary or similar

Charts: Chart.js or Recharts

Feature Breakdown

1. Authentication and User System

User Registration and Login

- Secure signup and login
- Password hashing using bcrypt
- JWT based authentication
- Protected routes on frontend and backend

User Profile

- Name, email, avatar upload
- Role in each workspace
- View assigned tasks across all boards

2. Workspaces (Top Level Organization)

A workspace represents a team or company.

Create Workspace

- Any registered user can create a workspace
- Creator becomes workspace admin

Invite Members

- Invite users via email
- Assigned role:
 - Admin, can manage workspace and members

- Member, can work on tasks

Workspace Dashboard

- Overview of:
 - Total tasks
 - Completed tasks
 - Tasks due soon
- Charts showing team productivity

3. Boards (Project Level)

Boards represent projects inside a workspace.

Board Management

- Create, rename, delete boards
- Each board belongs to one workspace
- Example boards:
 - Website Redesign
 - Mobile App Launch

4. Lists (Task Stages)

Lists are columns inside a board.

Examples:

- To Do
- In Progress
- Review
- Done

Features

- Create, rename, delete lists
- Drag and drop lists to reorder them

5. Tasks (Core of the App)

Tasks live inside lists.

Task Details

Each task contains:

- Title
- Description
- Due date
- Priority (Low, Medium, High)
- Assigned members
- Attachments
- Comments

Task Actions

- Create, edit, delete task
- Drag and drop task between lists
- Assign one or multiple members
- Add due date and get deadline alerts
- Upload files (images, docs, PDFs)

6. Comments and Collaboration

Each task has a discussion section.

- Members can post comments
- Show commenter name and time
- Realtime comment updates using Socket.io

This replaces messy email threads.

7. Realtime Updates

Using Socket.io:

- When a task moves to another list, everyone sees it instantly
- New comments appear live
- Task updates sync across all users on the board

This makes the app feel alive and collaborative.

8. Notifications System

Users get notified when:

- They are assigned to a task
- Someone comments on their task
- A due date is near
- They are invited to a workspace

Notifications can be:

- In app notification panel
- Optional email notifications

9. Activity Logs

Every important action is recorded:

- Task created
- Task moved
- Member assigned
- Due date changed

Admins can see who did what and when. This is very useful in real teams.

10. File Attachments

Users can upload files to tasks:

- Images
- PDFs
- Documents

Files are stored in cloud storage and linked to the task.

11. Dashboard and Analytics

Workspace dashboard shows:

- Tasks completed per week
- Tasks per member
- Overdue tasks
- Productivity charts

This gives managers insight into team performance.